

Disproof of Conjecture 1665: the proposed tight burning family

Local verified working submission
October 3, 2026

Claim being disproved. The conjecture states, among other claims, that Cartesian products of the Petersen graph with paths have burning number $\lceil \sqrt{n} \rceil + 1$, where n is the order of the product. We disprove this universal tight-family statement. We make no assertion about the separate proposed upper bound for arbitrary graphs.

Definitions and graph. Let P be the Petersen graph on vertices $0, \dots, 9$. Its edges are the outer cycle $i(i+1)$ for $i \in \mathbb{Z}/5\mathbb{Z}$, the inner cycle $(5+i)(5+i+2)$, and the five spokes $i(5+i)$. Let P_2 be the path with two vertices $0, 1$. Set $G = P \square P_2$. Write the product vertex (u, j) as the integer $u + 10j$. Thus G is a simple graph of order 20.

At each burning round, choose a vertex unburned at the beginning of the round, ignite it, and spread the older fires to all adjacent vertices. For the burned set B and source s , the update is

$$B \mapsto B \cup \{s\} \cup \{v : (\exists u \in B) uv \in E(G)\}.$$

The burning number is the least number of rounds in which every vertex can be burned.

A four-round certificate. Choose the successive sources $0, 1, 2, 12$. Direct application of the update rule gives

$$\begin{aligned} B_0 &= \emptyset, \\ B_1 &= \{0\}, \\ B_2 &= \{0, 1, 4, 5, 10\}, \\ B_3 &= \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 14, 15\}, \\ B_4 &= \{0, 1, \dots, 19\}. \end{aligned}$$

Each source is outside the burned set at the beginning of its round: $0 \notin B_0$, $1 \notin B_1$, $2 \notin B_2$, and $12 \notin B_3$. Consequently $b(G) \leq 4$. Equivalently, the Petersen graph has diameter two and its product with P_2 has diameter three, so one initial fire already reaches every vertex by the fourth round.

On the other hand, $4^2 < 20 \leq 5^2$, so $\lceil \sqrt{20} \rceil + 1 = 6$. Therefore

$$b(P \square P_2) \leq 4 < 6 = \lceil \sqrt{|V(P \square P_2)|} \rceil + 1.$$

This contradicts the claimed equality and disproves Conjecture 1665.

Lean verification and semantic correspondence. The self-contained file `Main.lean` imports only `Std`. The predicate `petPath m` is the Cartesian product with the m -vertex path. Theorems `product_encoding` and `product_encoding_bijective` verify that the 20 labels used here represent exactly $P \square P_2$, and `graph_simple` checks symmetry and absence of loops. The definitions `advance`, `run`, and `legal` implement precisely the update rule above, including the condition on new sources. Theorems `legal_sources` and `all_burned` check the complete finite process, not merely its length. The predicate `BurningNumberIs` asserts successful burning and minimality. Finally, `not_tight_family_claim` refutes the universally quantified equality for all products with nonempty paths. Its reported axiom dependency is only `propext`. No `sorry`, `admit`, `native_decide`, or additional axiom is used.

Reproduction. Run `lean Main.lean` with Lean 4.19.0. The command exits successfully and prints the axiom dependencies of the principal results. All finite certificates use kernel-checked `decide`.